
Solution Architecture

Date: 29 june 2026

Team ID: (xxxxxx)

Project Name: Human Development Index (HDI) Prediction Using Machine Learning

Maximum Marks: 5 Marks

Solution Architecture Diagram Content

Presentation / Client Layer

Technologies:

- HTML5
- CSS3
- Bootstrap
- JavaScript

Purpose:

Provides a user-friendly web interface where users enter the following details:

- Life Expectancy
- Mean Years of Schooling
- Expected Years of Schooling
- GNI Per Capita

Displays the predicted HDI category.

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API Gateway / Web Server

Technology:

- Flask (Python Web Framework) **Purpose:**
- Receives user requests.

- Sends input data to the machine learning model.
- Returns prediction results to the web page.

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Auth Service

Technology:

- Flask Session (Optional) **Purpose:**
 - Basic user authentication (if implemented).
 - Manages user sessions securely.
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Core Logic Service

Technologies:

- Python
 - Scikit-learn
 - Pandas
 - NumPy
 - Joblib **Purpose:**
 - Reads user input.
 - Performs data preprocessing.
 - Loads the trained machine learning model (model.pkl).
 - Predicts the HDI category.
 - Sends prediction results back to the Flask application.
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External API Integrations

Technology:

- No external APIs used.

Purpose:

The project works completely offline using a local dataset and trained machine learning model.

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Data / Storage Layer

Technologies:

- CSV Dataset
- Joblib (model.pkl)

Purpose:

Stores:

- HDI dataset
- Trained machine learning model
- Input data for prediction

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Collects user input and displays the predicted HDI category through a web interface.	HTML5, CSS3, Bootstrap, JavaScript
API Gateway	Handles HTTP requests, processes form data, and connects the frontend with the machine learning model.	Flask (Python)
Core Logic Service	Performs data preprocessing, loads the trained model, predicts the HDI category, and returns the prediction result.	Python, Scikit-learn, Pandas, NumPy, Joblib
Database / Storage	Stores the HDI dataset and the trained model for prediction.	CSV File, Joblib (model.pkl)

VS Code Project Structure HDI-

Prediction/

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|—— dataset/

| |—— hdi_dataset.csv

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|—— models/

| |—— model.pkl

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|—— static/

| |—— style.css

| |—— script.js

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|—— templates/

| |—— index.html

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|—— app.py

|—— train_model.py

|—— requirements.txt

|—— README.md

|—— .gitignore

Team Members

Team Member	Role	Responsibility
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Team Member	Role	Responsibility
G Naveen	Team Lead	Project planning, GitHub management, and project integration
Chandana Busireddy	Member	Dataset collection, preprocessing, and documentation
Gandham Joshna	Member	Data visualization and exploratory data analysis
Nandineni Gowthami	Member	Flask web application development and testing
Shaik Talha	Member	Machine learning model training, report preparation, GitHub updates, and final presentation

This content is tailored for your **HDI Prediction** project using **Python, Flask, Scikitlearn, and VS Code**, and is ready to be copied into your **Solution Architecture** reference document. Replace the placeholder **Team ID** with your actual team number before submission.